Day3

Task 1: Engineered Feature List

Six new features were built directly from existing columns, using only current-row data so no information leaks across rows or from the target. Each was checked against the target using mutual information or a grouped target rate before being added to the pipeline.

Feature	Type	Why it might help
age_bucket	categorical	captures the non-linear rise/plateau of income with age
hours_bucket	categorical	part-time, full-time, and overtime likely relate to income differently, not linearly
has_capital_gain	boolean	most rows have zero capital gain, so simply having any is a strong signal on its own
log_capital_gain	numeric	compresses the heavy right skew so large outliers do not dominate
higher_education	boolean	turns the useful Day 1 baseline rule into an explicit reusable feature
edu_hours_interaction	numeric	captures the combined effect of high education and long hours worked together

Task2: Pipeline and Cross-Validated Model Comparison

The engineered features were added through a FunctionTransformer placed before the existing imputation, scaling, and one hot encoding steps, so every model sees the same consistent preprocessing. Three models were compared with identical preprocessing using 5 fold StratifiedKFold cross-validation on the training set: Logistic Regression, Random Forest, and a gradient boosting model (HistGradientBoosting, a drop-in stand-in for XGBoost or LightGBM). Precision remained the primary metric, with ROC AUC and F1 tracked alongside it. Boxplots of the fold scores for all three metrics are in the notebook, showing both the average level and the spread across folds for each model.

Model	Precision (mean +/- std)	ROC AUC	F1
Logistic Regression	0.7433 +/- 0.0079	0.9123 +/- 0.0035	0.6701 +/- 0.0042
Random Forest	0.7198 +/- 0.0115	0.9016 +/- 0.0028	0.6630 +/- 0.0049
Gradient Boosting (HistGB)	0.7764 +/- 0.0116	0.9259 +/- 0.0038	0.7051 +/- 0.0076

Exact mean and standard deviation values are shown in the table above. The boosted model has the highest mean precision and ROC AUC, while Logistic Regression is second on precision and Random Forest is lower on both precision and ROC AUC.

Task3: Statistical Test Results

The top two models by mean precision were compared with a paired t-test across the 5 matched cross-validation folds, with a Wilcoxon signed-rank test run alongside it as a non-parametric cross-check. With only 5 paired folds this test has limited statistical power, so a p-value above 0.05 is read as not enough evidence of a real difference rather than proof there is no difference. When the p-value is small and the gap between means is large relative to their standard deviations, the better performing model is treated as genuinely stronger. When the p-value is large or the means nearly overlap, the simpler and cheaper model is preferred, since the added complexity is not buying a real improvement. The top two models by mean precision were Gradient Boosting (HistGB) and Logistic Regression. Paired t-test: $t = 7.4361$, $p = 0.0017$. Wilcoxon signed-rank test: statistic = 0.0, $p = 0.0625$.

Task 4: Feature Importance

Random Forest feature importances and Logistic Regression coefficients were both reviewed for the engineered features specifically. log_capital_gain, has_capital_gain, and edu_hours_interaction consistently ranked among the more useful engineered features for both models, which makes sense since capital gain was already the strongest Day 1 signal and the education-hours interaction captures a combined effect neither raw feature expresses alone. age_bucket and hours_bucket added smaller but still positive value, mainly by giving the linear model a way to represent non-linear patterns that raw age and raw hours cannot.

Task5: Feature Selection and Recommendation for Tomorrow

A SelectKBest step using mutual information was tested on top of the existing pipeline, keeping roughly half of the encoded columns, and compared against using all features on both cross-validated precision and training time. If the reduced feature set performs within a small margin of the full set while training faster, it is worth carrying forward into tomorrow's hyperparameter tuning, since fewer features mean faster tuning iterations without giving up real performance. If precision drops noticeably, the full engineered feature set is kept instead.

Recommended for tomorrow's hyperparameter tuning:

- Carry forward Gradient Boosting (HistGB) and Logistic Regression as the top 2 models by mean precision
- Keep log_capital_gain, has_capital_gain, and edu_hours_interaction regardless of the selection outcome, given their consistent signal
- Use the reduced feature set from SelectKBest only if its precision and speed trade-off looks favorable: full features (120 columns) achieved precision 0.7764 +/- 0.0116 in 14.0s, while selected features (k=60) achieved 0.7716 +/- 0.0130 in 94.1s. The full feature set is therefore preferable here.
- Re-check the paired statistical test once hyperparameters are tuned, since model rankings can shift after tuning